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GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO4

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 1

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

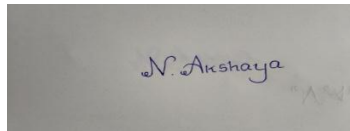
Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **N.AKSHAYA(192425129)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student:N.AKSHAYA

Criteria	Problem Understanding & Requirement Analysis (2)	Model Architecture & Design (2.5)	Application of Deep Learning Techniques (2.5)	Model Development & Implementation Strategy (2)	Performance Analysis & Justification (1)	Total (10)
Weightage	20 %	30%	25%	15 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS:05

Total Marks:25

1. Design and develop a model for a hospital has only 2,000 labeled X-ray images for classifying images into normal and abnormal categories. A pre-trained ResNet model is available. **Design a CNN-based transfer learning model** by identifying which layers should be frozen, which layer should be replaced, and how the model should be trained.

Answer:

The hospital has only 2,000 labeled X-ray images, so a pre-trained ResNet model is used to reduce overfitting and training time.

- Resize X-ray images to $224 \times 224 \times 3$ and normalize them.
- Use a pre-trained ResNet as the feature extractor.
- Freeze the initial ResNet layers because they contain general features such as edges and textures.
- Replace the original classification layer with:
 - Global Average Pooling
 - Dense layer with ReLU
 - Dropout layer
 - Dense layer with 2 Softmax outputs
- Initially train only the new classification layers.
- Fine-tune the final ResNet layers using a low learning rate.
- Use Adam optimizer and categorical cross-entropy loss.
- Evaluate using Accuracy, Precision, Recall, F1-score and Confusion Matrix.

Architecture:

X-Ray Image $\rightarrow$ ResNet $\rightarrow$ Global Average Pooling $\rightarrow$ Dense + ReLU $\rightarrow$ Dropout $\rightarrow$ Softmax $\rightarrow$ Normal/Abnormal

2. An agricultural organization wants to identify five types of plant diseases from leaf images. Since the dataset contains limited training samples, the organization wants to reuse features efficiently. **Design a DenseNet-based classification model** and explain how dense connections can help the model reuse features during classification.

Answer:

A pre-trained DenseNet model is used to classify five plant diseases from limited leaf-image samples.

- Resize leaf images to $224 \times 224 \times 3$ and normalize them.
- Use DenseNet as the feature extractor.
- Freeze most pre-trained layers initially.
- Dense connections allow each layer to receive feature maps from all preceding layers.
- This improves feature reuse, gradient flow and reduces redundant learning.
- Add Global Average Pooling followed by Dense, Dropout and Softmax layers.
- Use 5 Softmax outputs corresponding to the five diseases.
- Fine-tune selected upper layers using a low learning rate.
- Use Adam optimizer and categorical cross-entropy loss.
- Evaluate using Accuracy, Precision, Recall, F1-score and Confusion Matrix.

Architecture:

Leaf Image $\rightarrow$ DenseNet $\rightarrow$ Dense Blocks $\rightarrow$ Global Average Pooling $\rightarrow$ Dense $\rightarrow$ Dropout $\rightarrow$ Softmax (5 Classes)

3. A university wants to predict a student's future academic performance using their **weekly attendance, assignment marks, internal assessment marks, and previous examination scores** collected over several weeks. **Design an LSTM-based model** that can process this sequential information and predict the student's performance category.

Answer:

An LSTM model is designed to predict student performance using weekly sequential information.

- Input features include attendance, assignment marks, internal assessment marks and previous examination scores.
- Represent each week as a feature vector.
- Arrange the data as a sequence of weekly observations.
- Feed the sequence into an LSTM layer to learn temporal dependencies.
- Add Dropout to reduce overfitting.
- Add a Dense layer followed by a Softmax output layer.
- Predict performance categories such as Low, Medium and High.
- Use Adam optimizer and categorical cross-entropy loss.
- Evaluate using Accuracy, Precision, Recall, F1-score and Confusion Matrix.

Architecture:

Weekly Student Data → LSTM → Dropout → Dense → Softmax → Performance Category

4. An organization wants to classify customer reviews as **positive, negative, or neutral**. Since the meaning of a word may depend on both the words appearing before and after it, **design a Bidirectional RNN model** for this application. Identify the major processing stages from input text to final sentiment prediction.

Answer:

A Bidirectional RNN is designed to classify customer reviews into Positive, Negative and Neutral categories.

- Collect and clean customer reviews.
- Tokenize the text into numerical sequences.
- Apply padding to obtain equal-length sequences.
- Use an Embedding layer to convert tokens into word representations.
- Process the sequence using a Bidirectional RNN.
- The forward RNN captures information from previous words.
- The backward RNN captures information from following words.
- Combine both representations for sentiment classification.
- Use a Dense layer and Softmax output with three classes.
- Use Adam optimizer and categorical cross-entropy loss.
- Evaluate using Accuracy, Precision, Recall, F1-score and Confusion Matrix.

Architecture:

Review Text → Cleaning → Tokenization → Padding → Embedding → Bidirectional RNN → Dense → Softmax → Sentiment

5. A software company wants to develop a system that translates **English sentences into Hindi**. The length of the input and output sentences may differ, and the system needs to understand the complete input sentence before generating the translated sentence. **Design an**

Encoder–Decoder model and describe the role of the encoder, decoder, hidden representation, and output layer in the translation process.

Answer:

An Encoder–Decoder model is used to translate English sentences into Hindi when input and output lengths may differ.

- The English sentence is given to the Encoder.
- The Encoder processes the complete input sequence.
- The Encoder produces a hidden representation containing the important information of the sentence.
- The Decoder uses this hidden representation to generate the Hindi sentence sequentially.
- The Decoder predicts one Hindi word at a time.
- The Output layer uses Softmax to predict the next word.
- Use <START> and <END> tokens to identify the beginning and end of the output.
- Train the model using paired English-Hindi sentences.
- Use Adam optimizer and categorical cross-entropy loss.
- Evaluate translation quality using BLEU score and validation loss.

Architecture:

English Sentence → Encoder → Hidden Representation → Decoder → Softmax → Hindi Sentence

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Requirement Analysis	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Model Architecture & Design	30%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Application of Deep Learning Techniques	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Model Development & Implementation Strategy	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Performance Analysis & Justification	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis